

Course on Probabilistic Design of Coastal Structures

Statistical Modelling of Environmental Conditions

01. DAILY WEATHER TYPES (DWT)

Jared Ortiz-Angulo Cantos

Researcher, GeoOcean Group, Universidad de Cantabria (UC)

ProCoast 2025
October 20th-23th, 2025

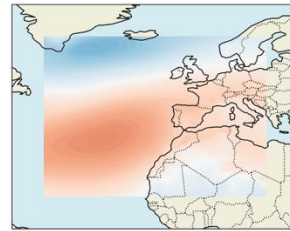


Framework

Climate-based Emulator

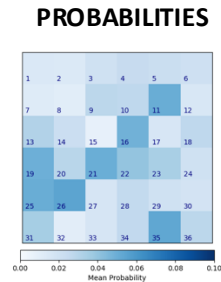
1. Predictors

Variables: SLP, SLP Gradient & 500 hPa Geo
Domain: North Atlantic Ocean & Mediterranean Sea
Data: ERA 5 reanalysis
Time series: 1940 to 2023 at daily scale
Spatial resolution: 1°

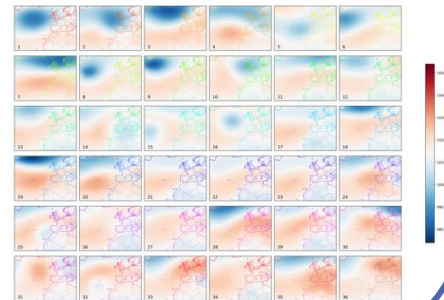


PCA + K-Means → 36 DWTs

KEEP THE PCs
EXPLAINING 95%
OF THE VARIANCE



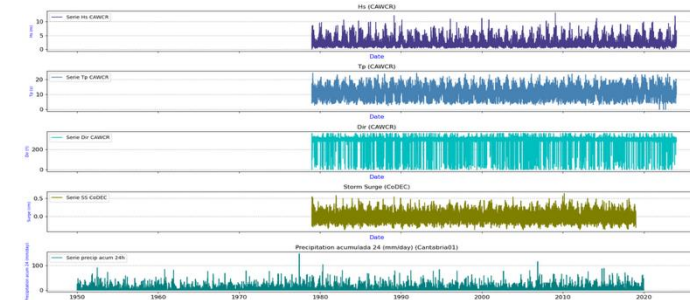
Camus et al., (2014)



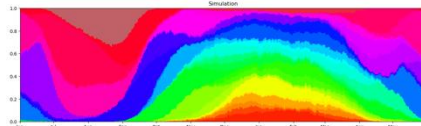
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Wave Data: CAWCR – hourly data (1979-2023)

Storm Surge: CoDEC – hourly data (1979 - 2023)

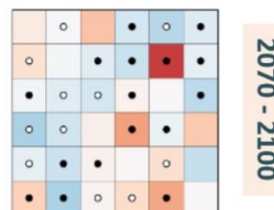


3. Seasonality Chronology



4. Large scale Climate patterns
relationship

5. Climate Change



6. Additional explanatory variables: Wind

7. Climate-based Emulator of met-ocean parameters

$$f(Hs, Tp, Dir, SS, precip) / DWTs$$

DATA

Predictor: **ERA5** (1940 - 2023)

<https://cds.climate.copernicus.eu/cdsapp#!/search?type=dataset>

P, W: ERA5 (Hersbach et al., 2020) – 0.25°/0.50°; **hourly**

1979-presente:

SSR

TWL

10mu

10mv

Significant wave height (swh)

Mean wave direction (mwd)

Mean wave period (mwp)

meanwp

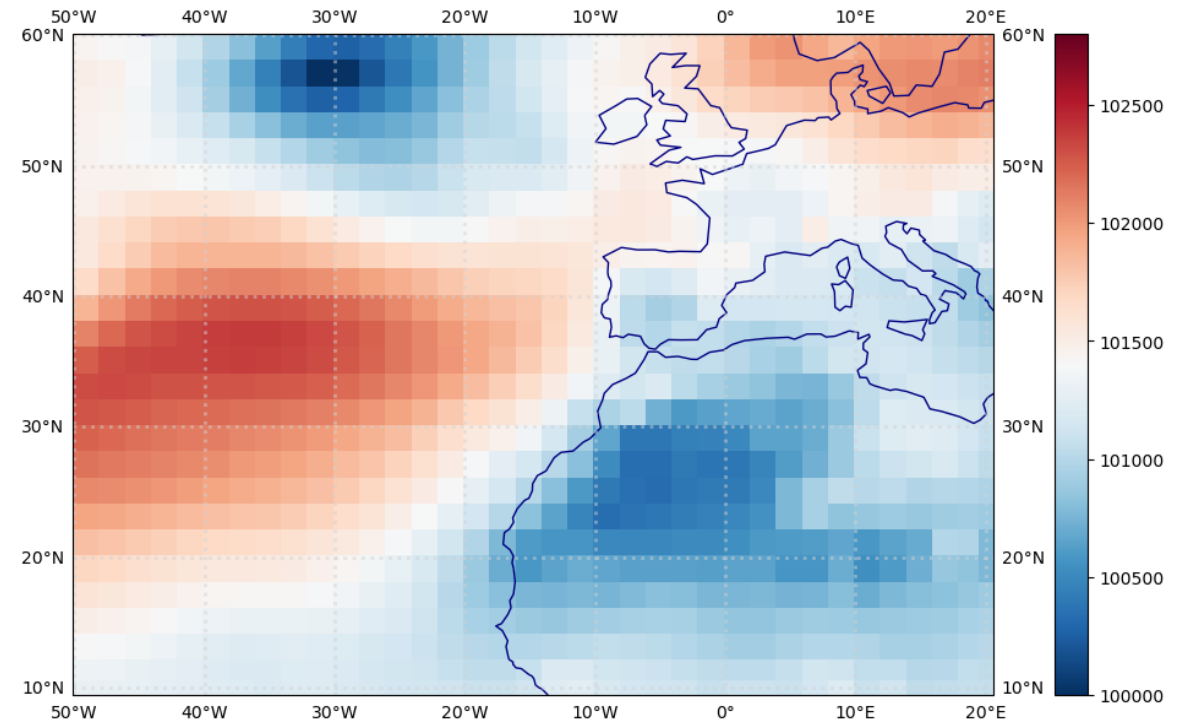
modelwp

modelb

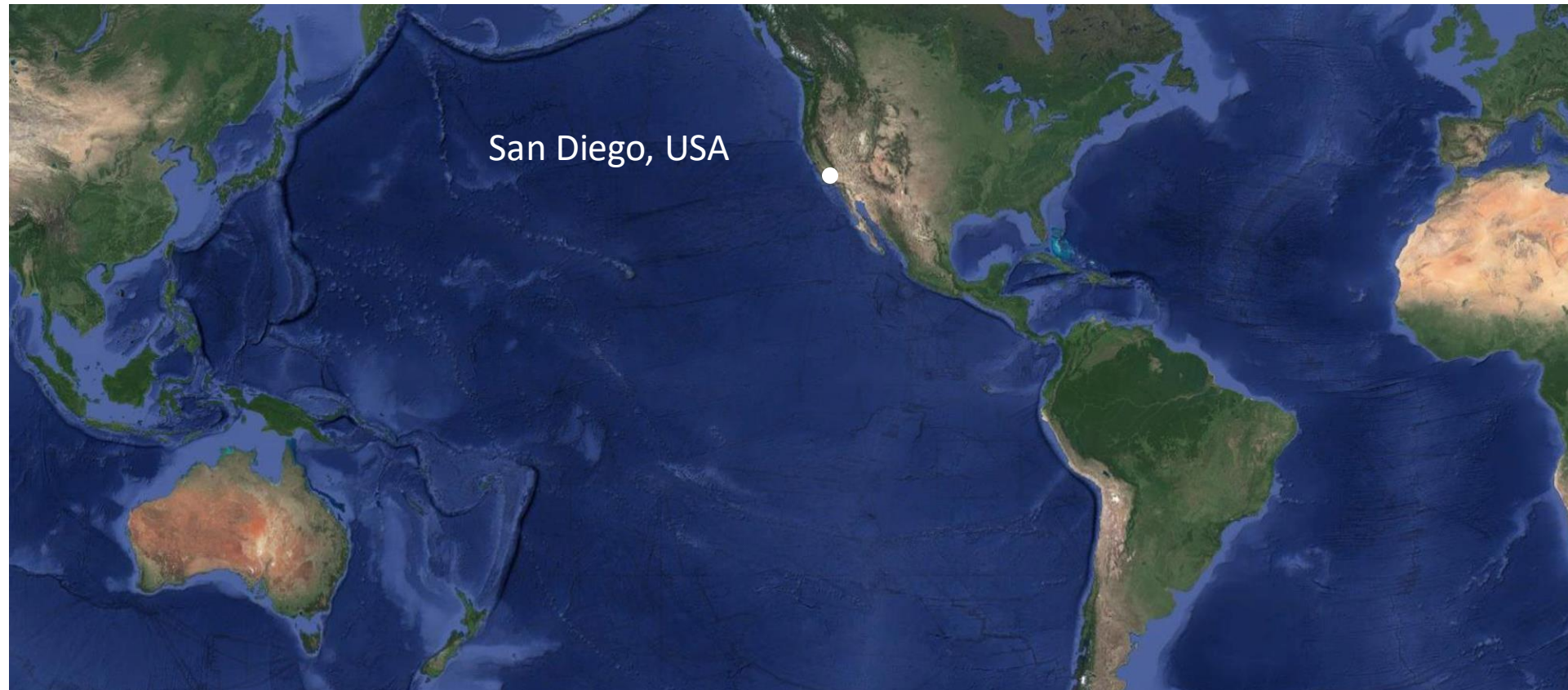
mslp

Wavewind

precipitation



DATA



Anderson et al 2019

ESTELA: a method for evaluating the source and travel time of the wave energy reaching a local area

Jorge Pérez · Fernando J. Méndez · Melisa Menéndez ·
Inigo J. Losada

Received: 26 January 2014 / Accepted: 4 June 2014
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Abstract The description of wave climate at a local scale is of paramount importance for offshore and coastal engineering applications. Conditions influencing wave characteristics at a specific location cannot, however, be fully understood by studying only local information. It is necessary to take into account the dynamics of the ocean surface over a large “upstream” wave generation area. The goal of this work is to provide a methodology to easily characterize the area of influence of any particular ocean location worldwide. Moreover, the developed method is able to characterize the wave energy and travel time in that area. The method is based on a global scale analysis using both geographically and physically based criteria. The geographic criteria rely on the assumption that deep water waves travel along great circle paths. This limits the area of influence by neglecting energy that cannot reach a target point, as its path is blocked by land. The individual spectral partitions from a global wave reanalysis are used to reconstruct the spectral information and apply the physically based criteria. The criteria are based on the selection of the fraction of energy that travels towards the target point for each analysed grid point. The method has been tested on several locations worldwide. Results provide maps that inform about the relative importance of different oceanic areas to the local wave climate at any target point. This information cannot be inferred from local parameters and agrees with information from other approaches. The

methodology may be useful in a number of applications, such as statistical downscaling, storm tracking and grid definition in numerical modelling.

Keywords Dynamical downscaling · Propagation · Spectrum · Statistical downscaling · Wave climate · Wave energy flux · Wave modelling

1 Introduction

Wave characteristics at a local scale play a major role in a wide range of issues (e.g. maritime works, navigation routes, offshore structures design, harbour operability, ecosystem distribution and flooding risk). These conditions are dominated by locally generated waves (wind sea) and waves from distant storms (swells). Hence, the need for accurate wave data has long encouraged studies on wave generation and propagation of global ocean swells. Studies by Munk (1947), Munk et al. (1963), Barber and Ursell (1948) and Snodgrass et al. (1966) contributed significantly to the scientific basis and introduced some of the currently accepted paradigms, e.g. great circle propagation and small dissipation of swells. Unfortunately, these earlier studies were limited by the availability of in situ measurements, i.e. a small number of stations concentrated in coastal areas and covering a short time period with long gaps.

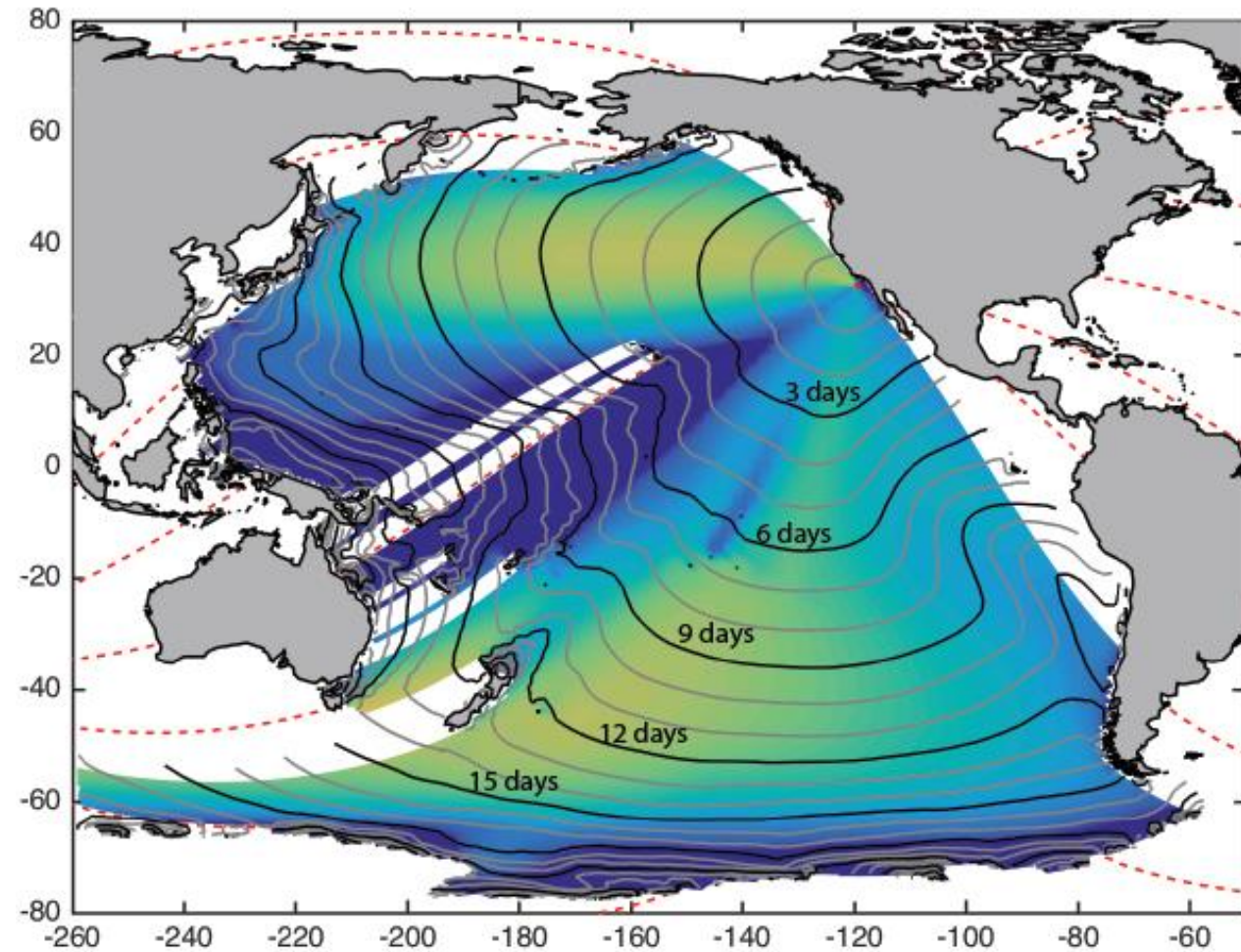
More recent studies (e.g. Collard et al. 2009) take advantage of the global coverage offered by Synthetic Aperture Radar data but they face similar constraints due to the coarse spatial and temporal resolution of these data. Despite the data limitation, studies on space and time variability of wave climate have lead to a huge improvement in the physics of numerical wave models. The improved physics and the availability of higher computational power have resulted in high-quality wave hindcast databases (e.g. Rascle et al. 2008; Reguero et al. 2012). The homogeneous and continuous data

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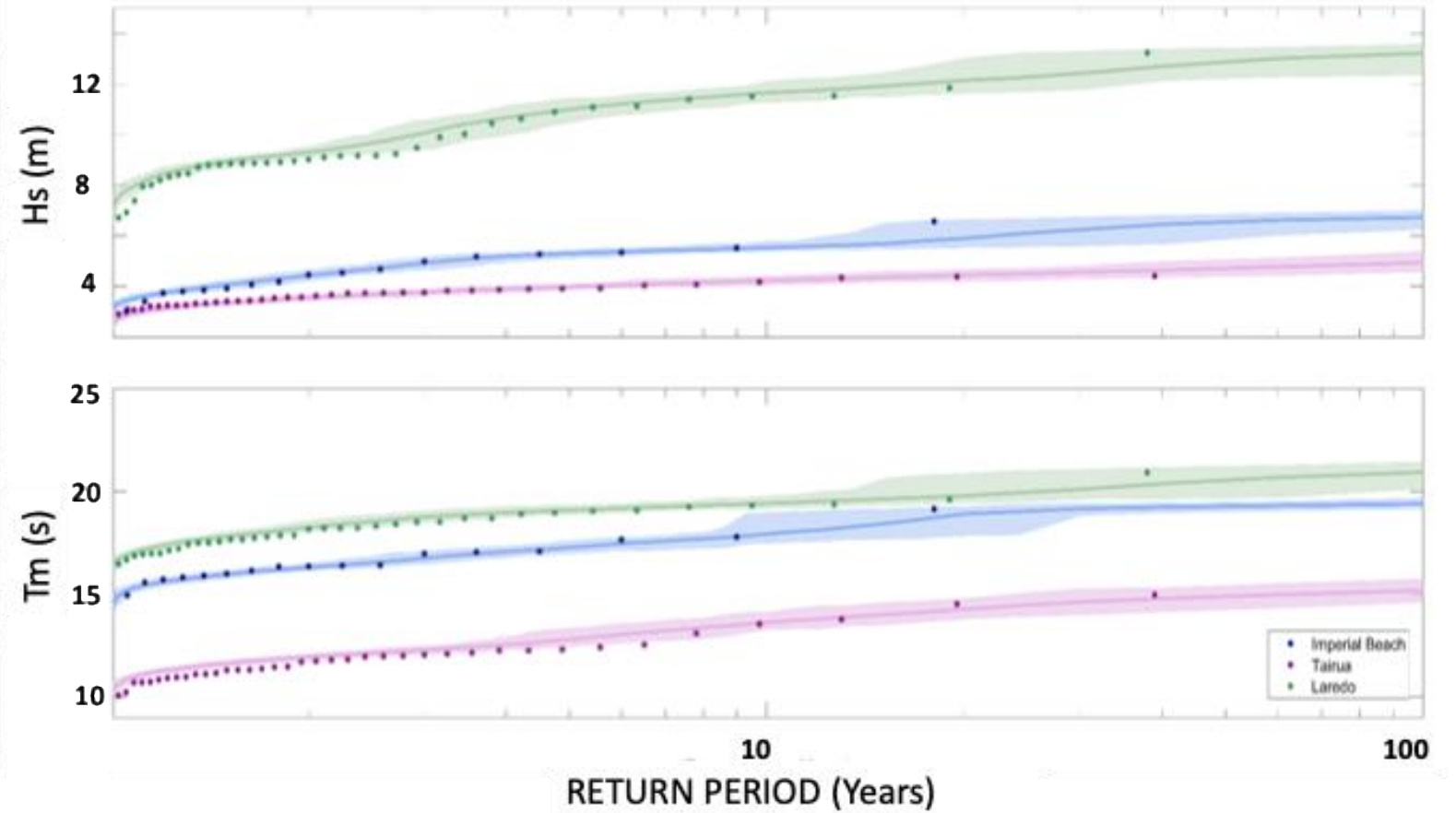
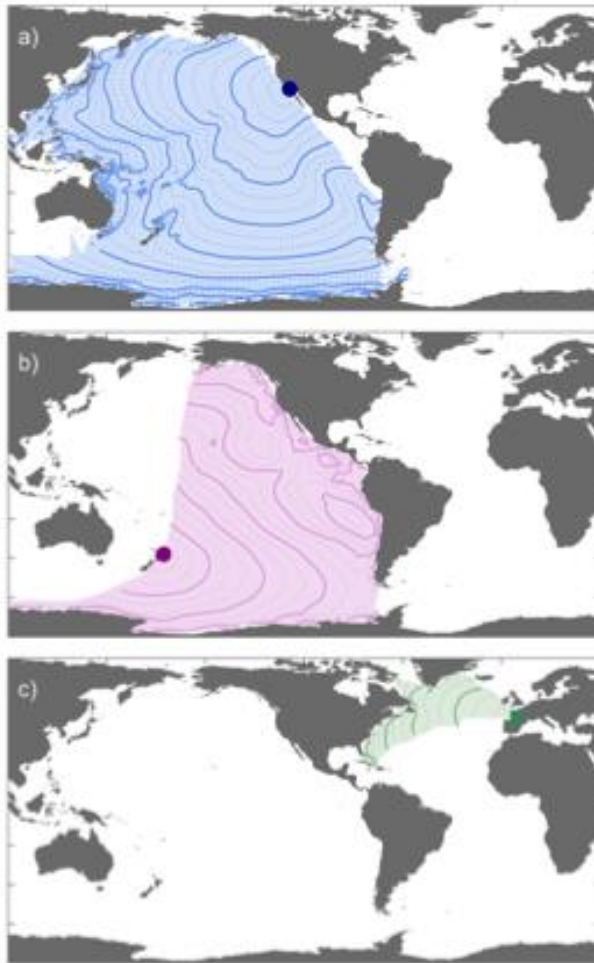
This article is part of the Topical Collection on the 13th International Workshop on Wave Hindcasting and Forecasting in Banff, Alberta, Canada October 27 – November 1, 2013

J. Pérez · F. J. Méndez (✉) · M. Menéndez · I. J. Losada
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e-mail: menendez@unican.es

Published online: 18 June 2014



DATA

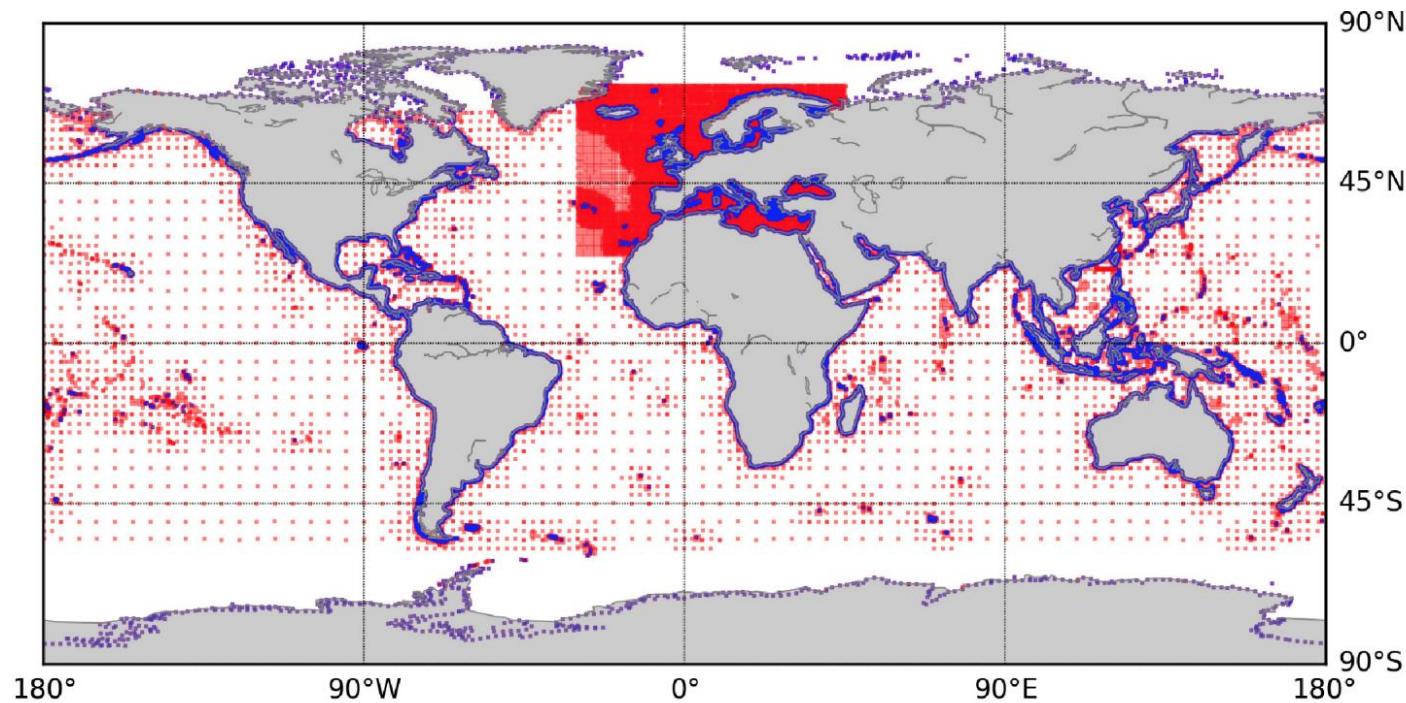


DATA

Predictand: CoDEC (1979 - 2023)

Surge & twl: CoDEC (Muis et al., 2020)

1.25 km/ 5 km along the coast; hourly data



Muis et al., 2020

frontiers
in Marine Science

ORIGINAL RESEARCH
published: 29 April 2020
doi: 10.3389/fmars.2020.00263



A High-Resolution Global Dataset of Extreme Sea Levels, Tides, and Storm Surges, Including Future Projections

Sanne Muis^{1,2*}, Maaijen Irazoqui Apecechea¹, Job Dullaart², Joao de Lima Rego³, Kristine Skovgaard Madsen⁴, Jian Su⁵, Kun Yan⁶ and Martin Verlaan^{1,4}

¹ Deltares, Delft, Netherlands, ² Institute for Environmental Studies (IVM), Vrije Universiteit Amsterdam, Amsterdam, Netherlands, ³ Danish Meteorological Institute, Copenhagen, Denmark, ⁴ Delft University of Technology, Delft, Netherlands

The world's coastal areas are increasingly at risk of coastal flooding due to sea-level rise (SLR). We present a novel global dataset of extreme sea levels, the Coastal Dataset for the Evaluation of Climate Impact (CoDEC), which can be used to accurately map the impact of climate change on coastal regions around the world. The third generation Global Tide and Surge Model (GTSM), with a coastal resolution of 2.5 km (1.25 km in Europe), was used to simulate extreme sea levels for the ERA5 climate reanalysis from 1979 to 2017, as well as for future climate scenarios from 2040 to 2100. The validation against observed sea levels demonstrated a good performance, and the annual maxima had a mean bias (MB) of -0.04 m, which is 50% lower than the MB of the previous GTSR dataset. By the end of the century (2071–2100), it is projected that the 1 in 10-year water levels will have increased 0.34 m on average for RCP4.5, while some locations may experience increases of up to 0.5 m. The change in return levels is largely driven by SLR, although at some locations changes in storms surges and interaction with tides amplify the impact of SLR with changes up to 0.2 m. By presenting an application of the CoDEC dataset to the city of Copenhagen, we demonstrate how climate impact indicators derived from simulation can contribute to an understanding of climate impact on a local scale. Moreover, the CoDEC output locations are designed to be used as boundary conditions for regional models, and we envisage that they will be used for dynamic downscaling.

Keywords: climate change, global model, extreme sea levels, sea-level rise, coastal flooding

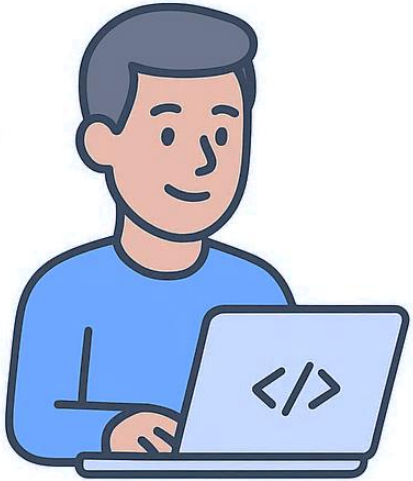
INTRODUCTION

The world's coastal areas are increasingly at risk of coastal flooding due to sea-level rise (SLR). By 2100, global mean sea-levels are projected to be 30–60 cm higher than today, even with a sharp reduction of greenhouse gas emissions (Church et al., 2013; Oppenheimer et al., 2019). In regions with little sea level variability, even an SLR of 10 cm could double the flooding frequency (Vitsardis et al., 2017). Without mitigation and adaptation, the expected economic annual losses due to flooding may come close to 10% of the global gross domestic product (Hinkel et al., 2014). Raising flood defenses is therefore critical in densely populated and economically important coastal areas, and is cost effective for 13% of the global coastline (Lindner and Hinkel, 2018). The stabilization of global temperatures to 1.5°C, as agreed upon in the Paris Agreement, will lead to a strong reduction of the economic impact of SLR (Nicholls et al., 2018).

DATA

How to Download CoDEC data?

GeoOcean thredds



Skilled
user

BlueMath / climate_services / probabilistic / MUSCLE_procoast25 / practice / Download_CoDEC_Data 1.ipynb

jaredoac "uploads"

825802f · 4 days ago History

Preview Code Blame 3980 lines (3980 loc) · 434 KB

Raw Download Edit

```
In [ ]: import sys

import os
import os.path as op
import xarray as xr

import matplotlib.pyplot as plt
from cartopy import crs as ccrs
from bluemath.plotting.common import load_coast
import numpy as np

import warnings
warnings.simplefilter(action='ignore', category=DeprecationWarning)

import numpy as np
import pandas as pd

import datetime
from datetime import timedelta
```

CoDEC

```
In [36]: longs = xr.open_dataset("https://geocean.sci.unican.es/thredds/dodsC/fagggregatedCODEC.nc").station_x_coordinate.
longs
```

```
Out[36]: array([ 24.089,  24.089,  26.902, ..., 139.146,  37.749,  37.954],
              shape=(43119,))
```


Predictor: **ERA5** (1940 - 2023)

<https://cds.climate.copernicus.eu/cdsapp#!/search?type=dataset>

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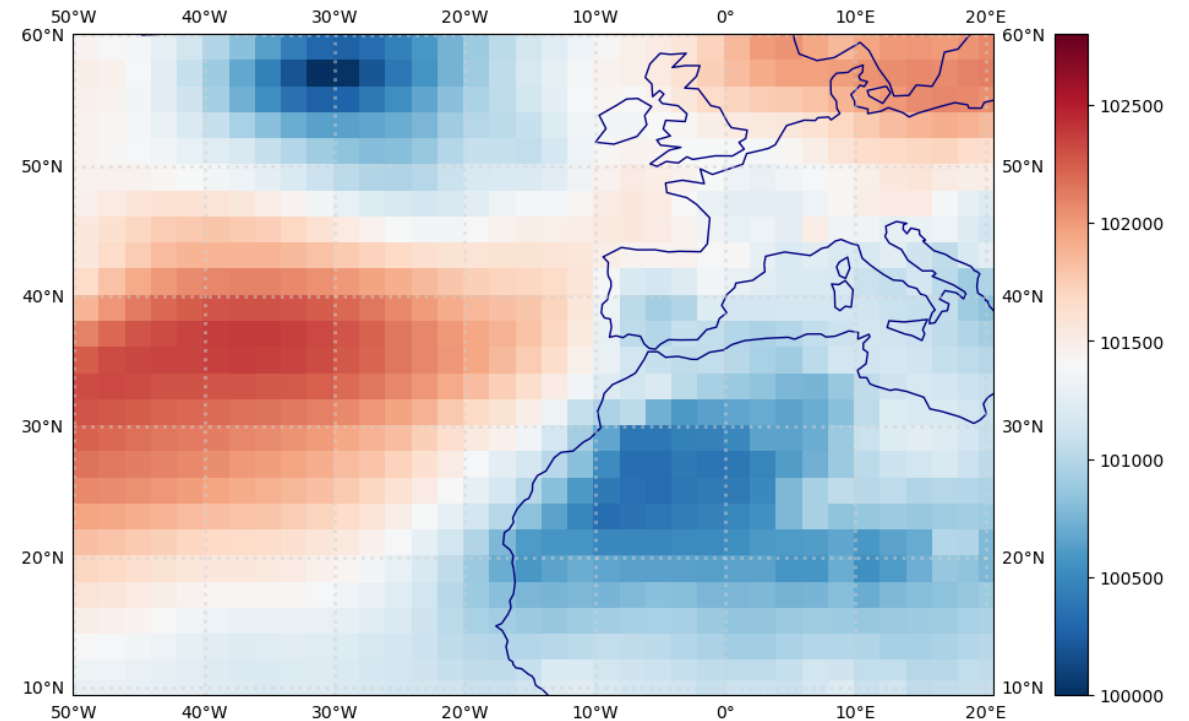
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Wavewind

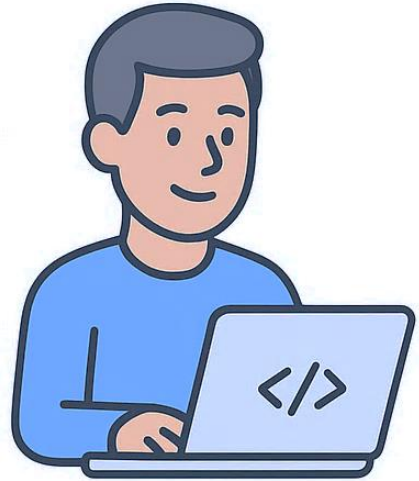
precipitation



DATA

How to Download ERA5 data?

DestinE (Destination Earth)



Skilled
user



[BlueMath](#) / [climate_services](#) / [probabilistic](#) / [MUSCLE_procoast25](#) / [practice](#) / [Download_ERA5_Data 1.ipynb](#)

jaredoac "uploads" 825802f · 4 days ago History

Preview Code Blame 2375 lines (2375 loc) · 327 KB

```
In [1]: import os
import os.path as op
import xarray as xr

import matplotlib.pyplot as plt
from cartopy import crs as ccrs
import numpy as np

p_db = op.join(os.getcwd(),

ERA5

In [ ]: PAT = "your_personal_access_token"

In [14]: # PRESSURE AT SINGLE LEVEL

In [ ]: # 3-4 mins

# Oporto
lon = 350.9
lat = 41.15

ds = xr.open_dataset(
    f"https://edh:{PAT}@data.earthdatahub.destine.eu/era5/reanalysis-era5-single-levels-ocean-v0.zarr",
    chunks={},
    engine="zarr",
).sel(longitude=300, latitude=41.15, method="nearest")
ds[["swl", "ppld", "mwd"]].load()
ds
```

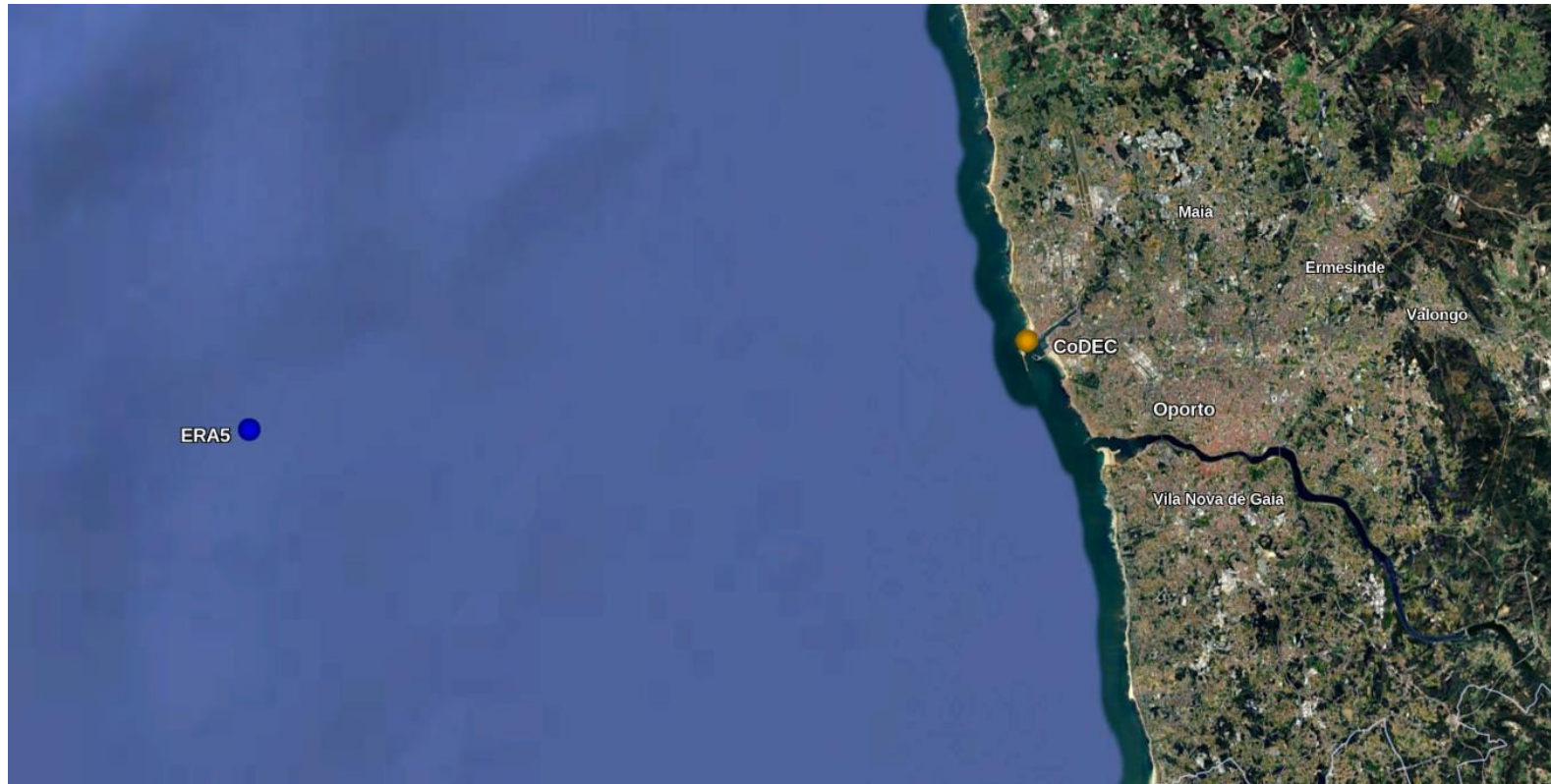
STUDY SITE

Study site: **Oporto (Portugal)**

Predictands:

Waves (Hs, Tp, Dir): **ERA5** – hourly data (1940-2025)

Storm Surge: **CoDEC** – hourly data (1979 - 2023)

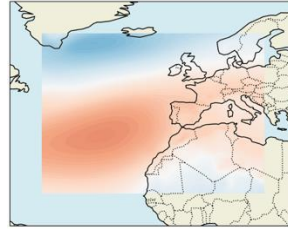


Framework

Climate-based Emulator

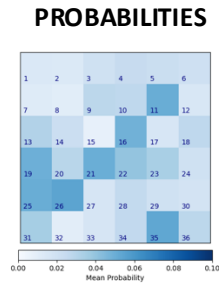
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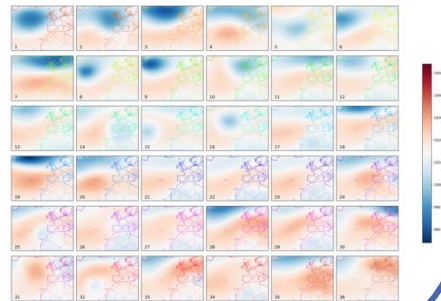


PCA + K-Means → 36 DWTs

KEEP THE PCs
EXPLAINING 95%
OF THE VARIANCE



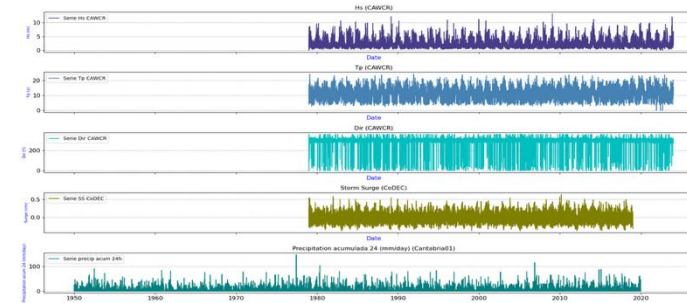
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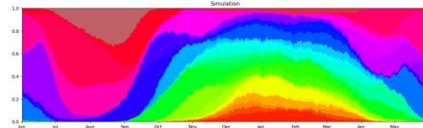
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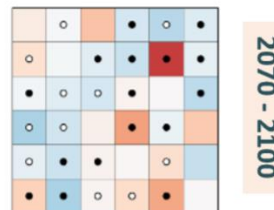


3. Seasonality Chronology



4. Large scale Climate patterns
relationship

5. Climate Change



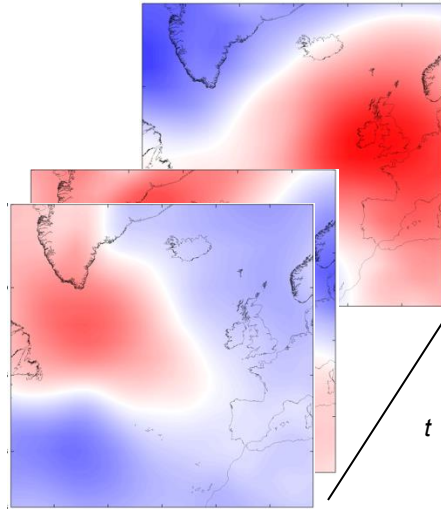
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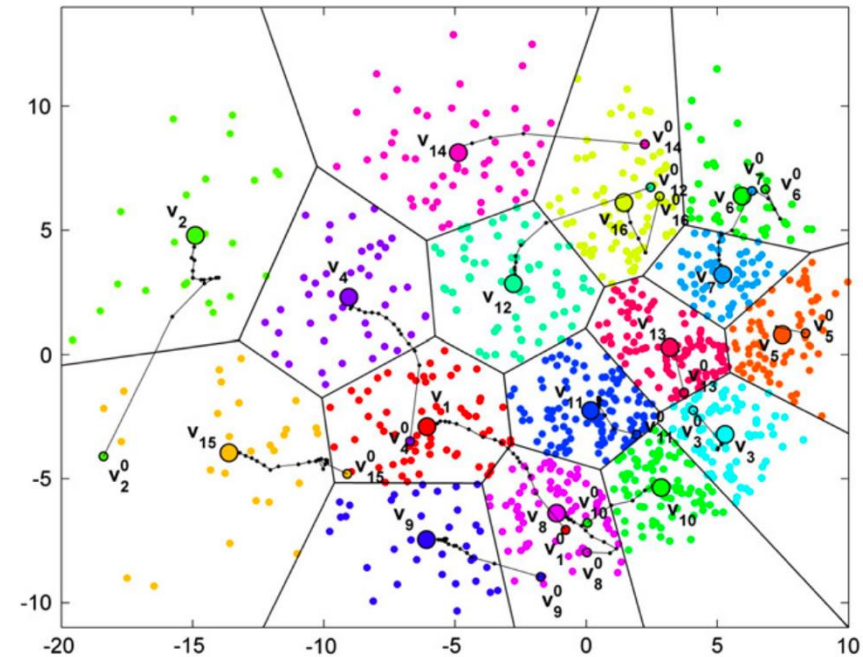
$$f(Hs, Tp, Dir, SS, precip) / DWTs$$

STATISTICAL TECHNIQUES

PRINCIPAL COMPONENT ANALYSIS: DIMENSIONALITY REDUCTION TECHNIQUE



K-MEANS: CLUSTERING ALGORITHM



PRINCIPAL COMPONENT ANALYSIS: DIMENSIONALITY REDUCTION TECHNIQUE

- PCA is a standard technique to eliminate redundant information with minimum loss of variability.
- It is also a data compression technique.
- Concept: projection of the data in a new space of lower dimension than the original one:

Sample of N n-dimensional observations:

$$X = \begin{pmatrix} x_{11} & \dots & x_{k1} & \dots & x_{N1} \\ \cdot & & \cdot & & \cdot \\ \cdot & & \cdot & & \cdot \\ \cdot & & \cdot & & \cdot \\ x_{1n} & \dots & x_{kn} & \dots & x_{Nn} \end{pmatrix}$$

$x_k = (x_{k1}, \dots, x_{kn})^T \quad k = 1, \dots, N$

From n dimensions to d

$d < n$

$$X = \sum_{i=1}^n e_i x_i \approx \bar{X} = \sum_{i=1}^d f_i \bar{x}_i$$

$\downarrow$
 $\{e_1, \dots, e_n\}$

$\swarrow$
 $\{f_1, \dots, f_d\}$

PCA

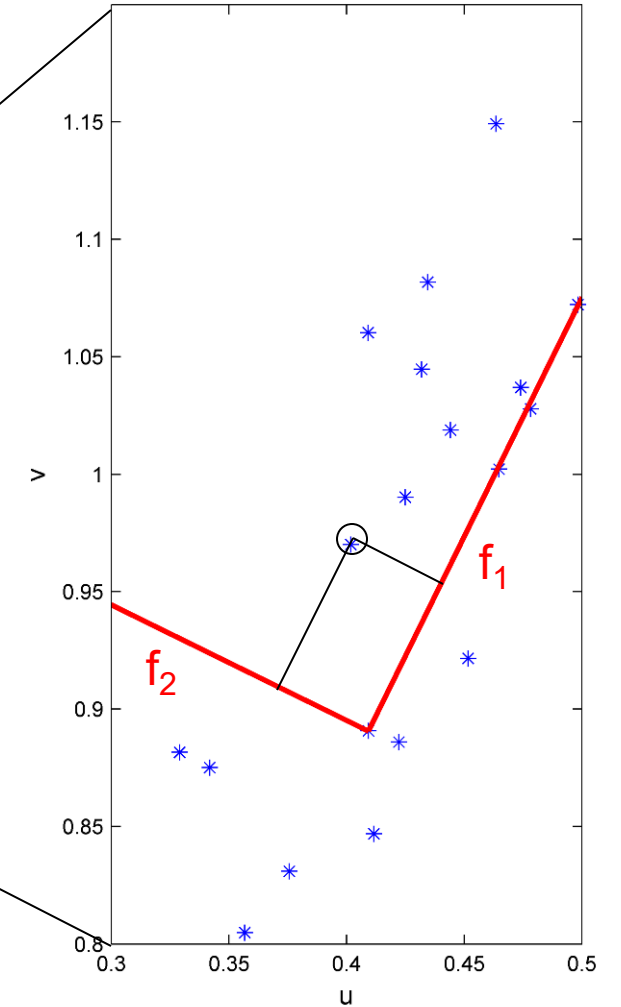
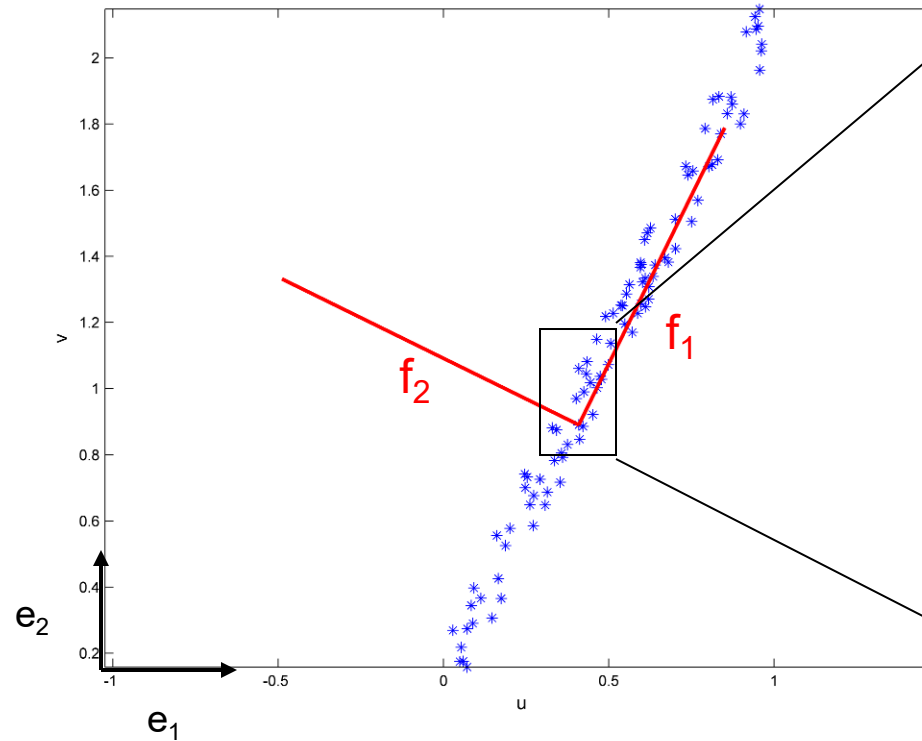
EOF: Empirical Orthogonal Function

Two-dimensional data sample

-The sample projected onto the new projection space will explain the maximum possible variance

-New vectors oriented in such a way that the sum of the squares of the projections of the data on these new vectors is maximal

EOF	% variance explained
1	94.7
2	5.3



PCA

Three-dimensional data sample

EOF	% variance explained
1	71.8
2	26.2
3	2.0

-Each vector of the new projection is defined as a linear combination of the vectors of the original.

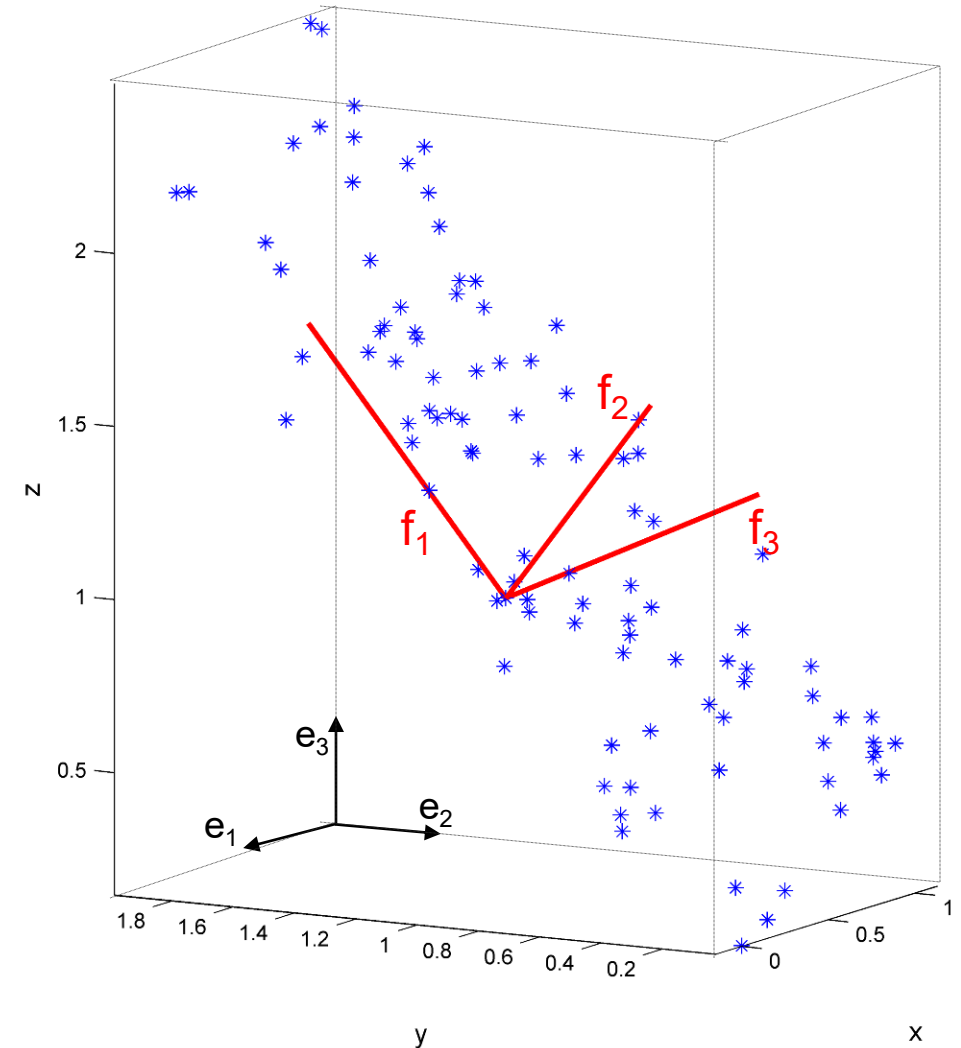
$$f_1 = 0.076e_1 + 0.687e_2 + 0.723e_3$$

$$f_2 = 0.938e_1 + 0.295e_2 + 0.182e_3$$

$$f_3 = 0.338e_1 + 0.664e_2 + 0.666e_3$$

$$f_1 = (0.076, 0.687, 0.723)$$

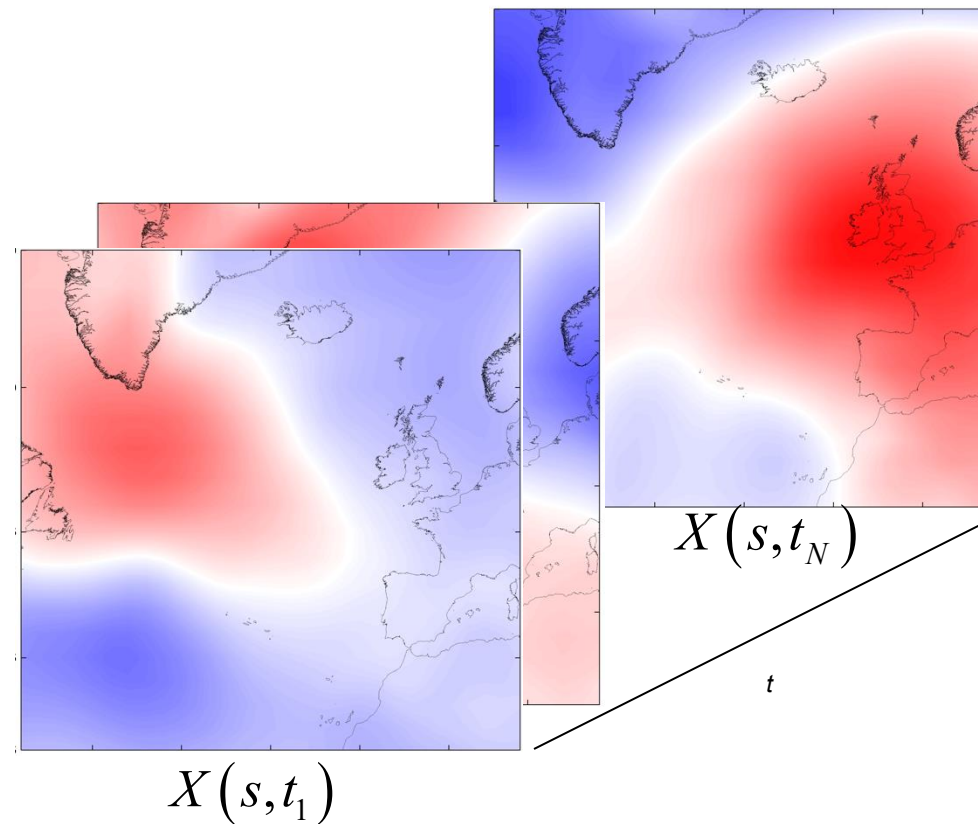
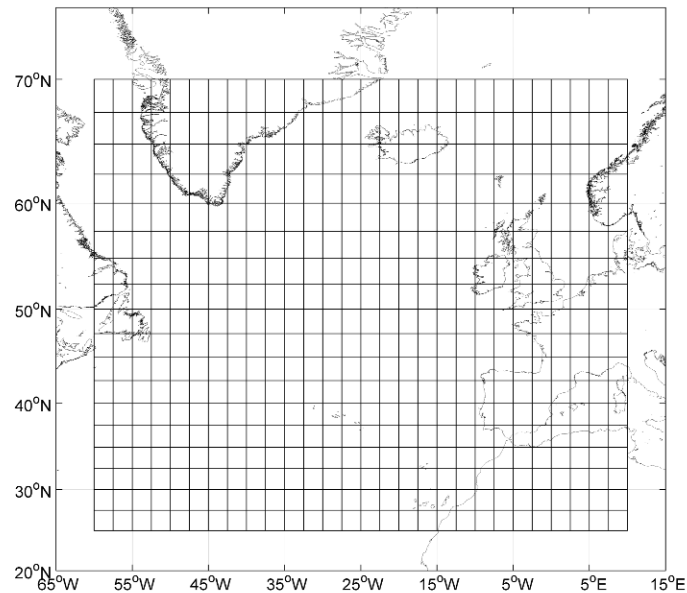
$$f_2 = (0.938, 0.295, 0.182)$$



PCA

Example Multidimensional data sample

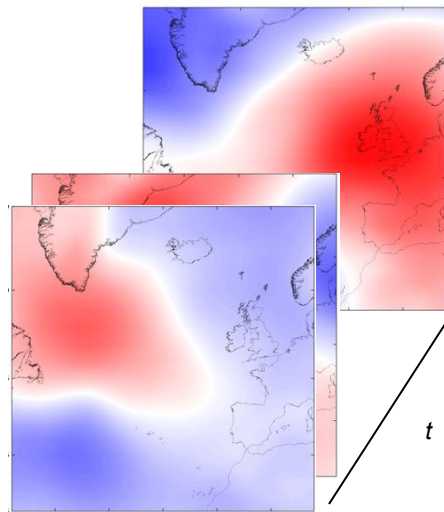
Time series of monthly pressures



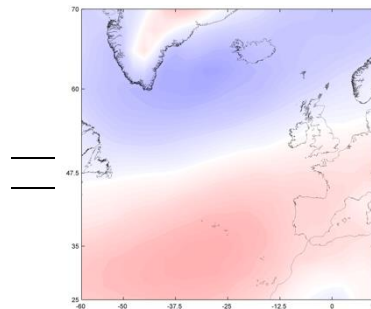
PCA

$$X(s, t_i) = \overline{X(s)} + EOF_1(s) \cdot PC_1(t_i) + EOF_2(s) \cdot PC_2(t_i) + \dots + EOF_d(s) \cdot PC_d(t_i)$$

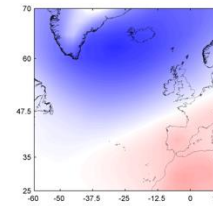
Example



Mean

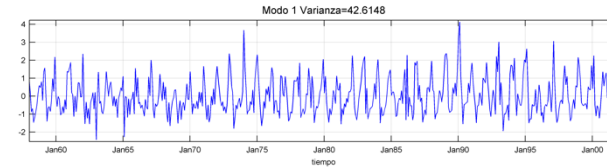


EOFs (Spatial modes)

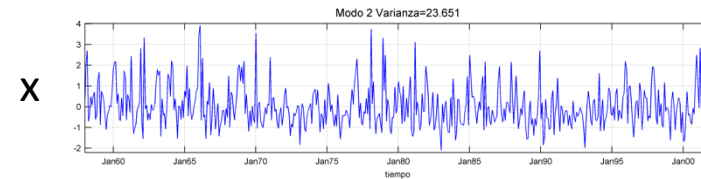


x

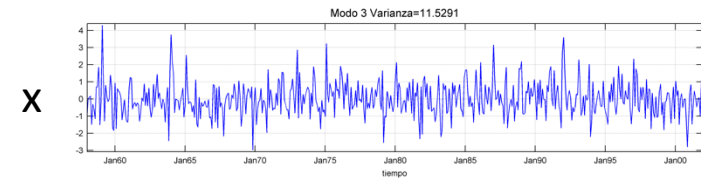
PCA (temporal coefficients)
Projections of data on EOFs



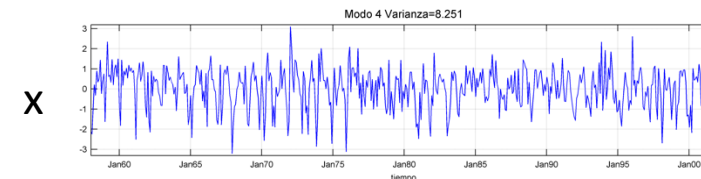
x



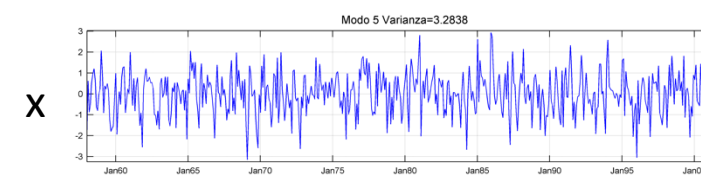
x



x



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PCA

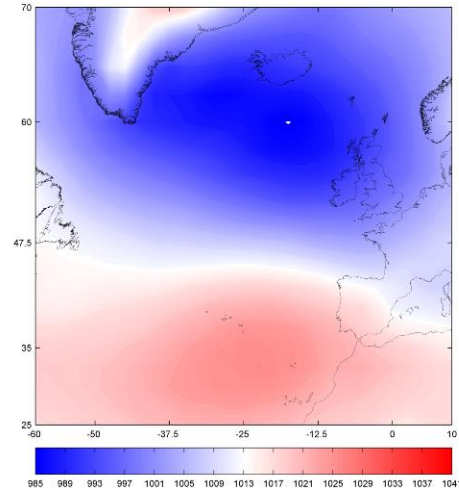
$$X(s, t_i) = \overline{X(s)} + EOF_1(s) \cdot PC_1(t_i) + EOF_2(s) \cdot PC_2(t_i) + \dots + EOF_d(s) \cdot PC_d(t_i)$$

Example

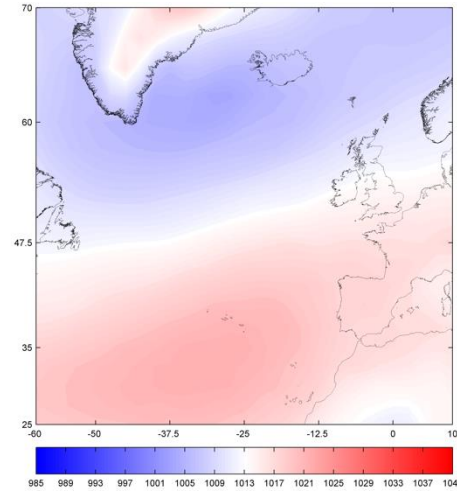
Pressure: December 1959

Mean

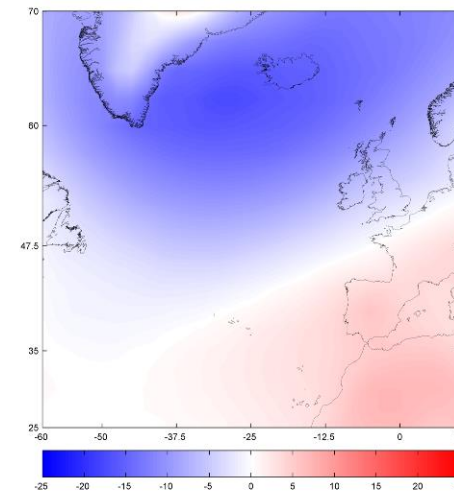
$EOF_1(s) \cdot PC_1(t_{Dec59})$



=

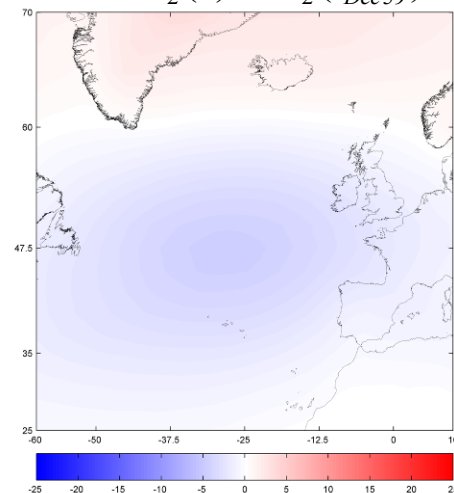


+



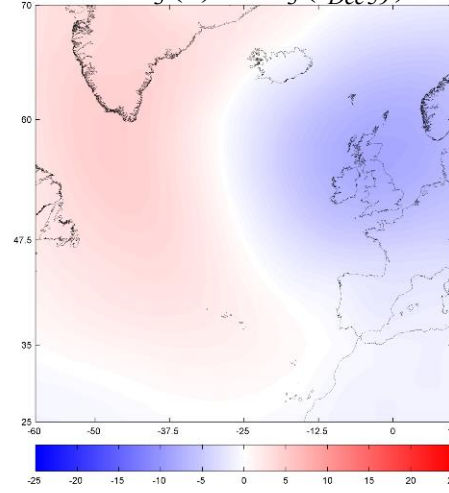
+

$EOF_2(s) \cdot PC_2(t_{Dec59})$



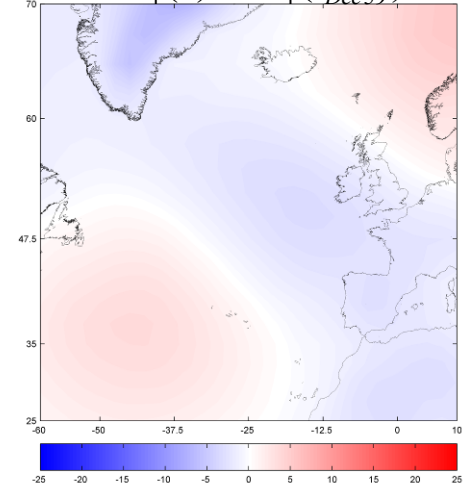
+

$EOF_3(s) \cdot PC_3(t_{Dec59})$



+

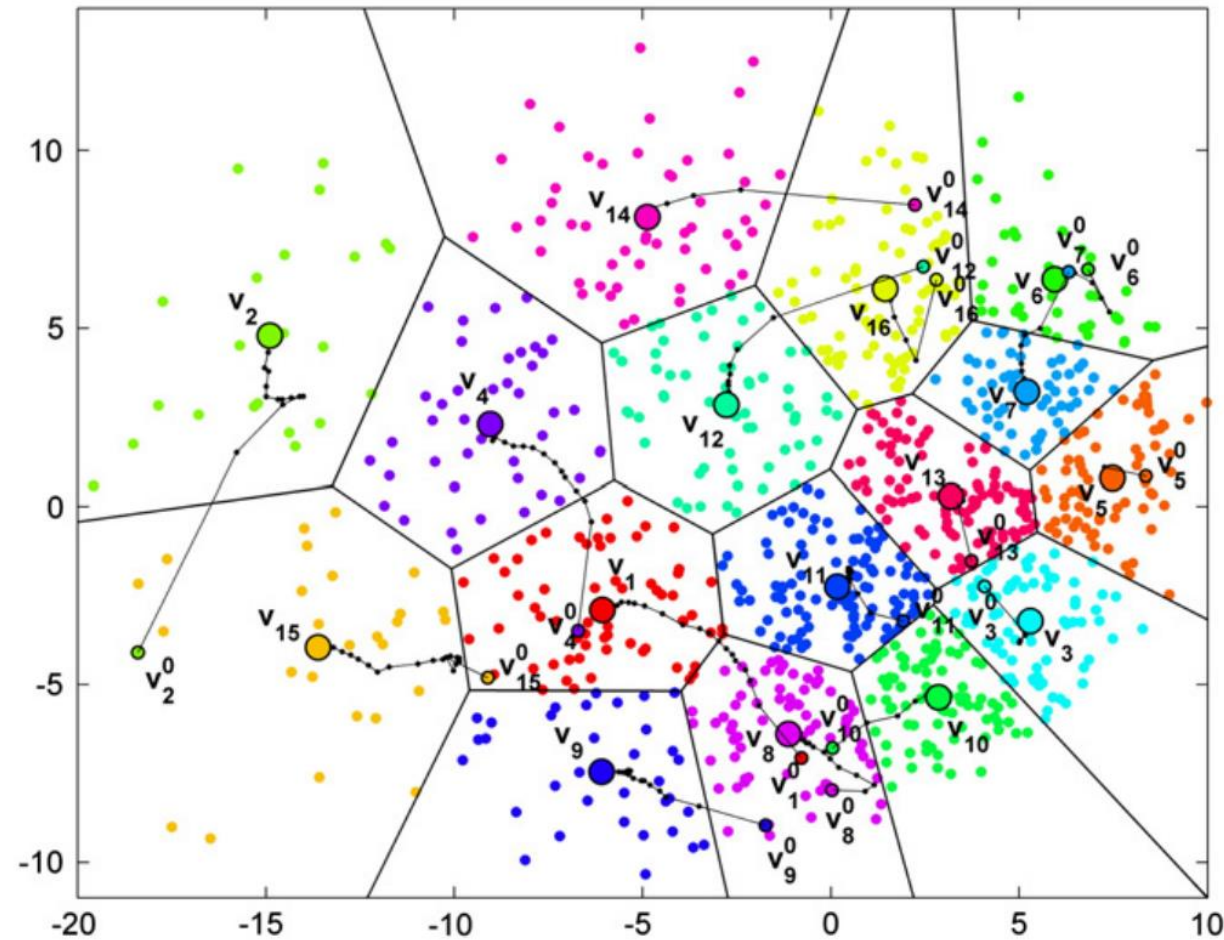
$EOF_4(s) \cdot PC_4(t_{Dec59})$



K-MEANS

CLUSTERING ALGORITHM

K-MEANS



Camus et al., 2011

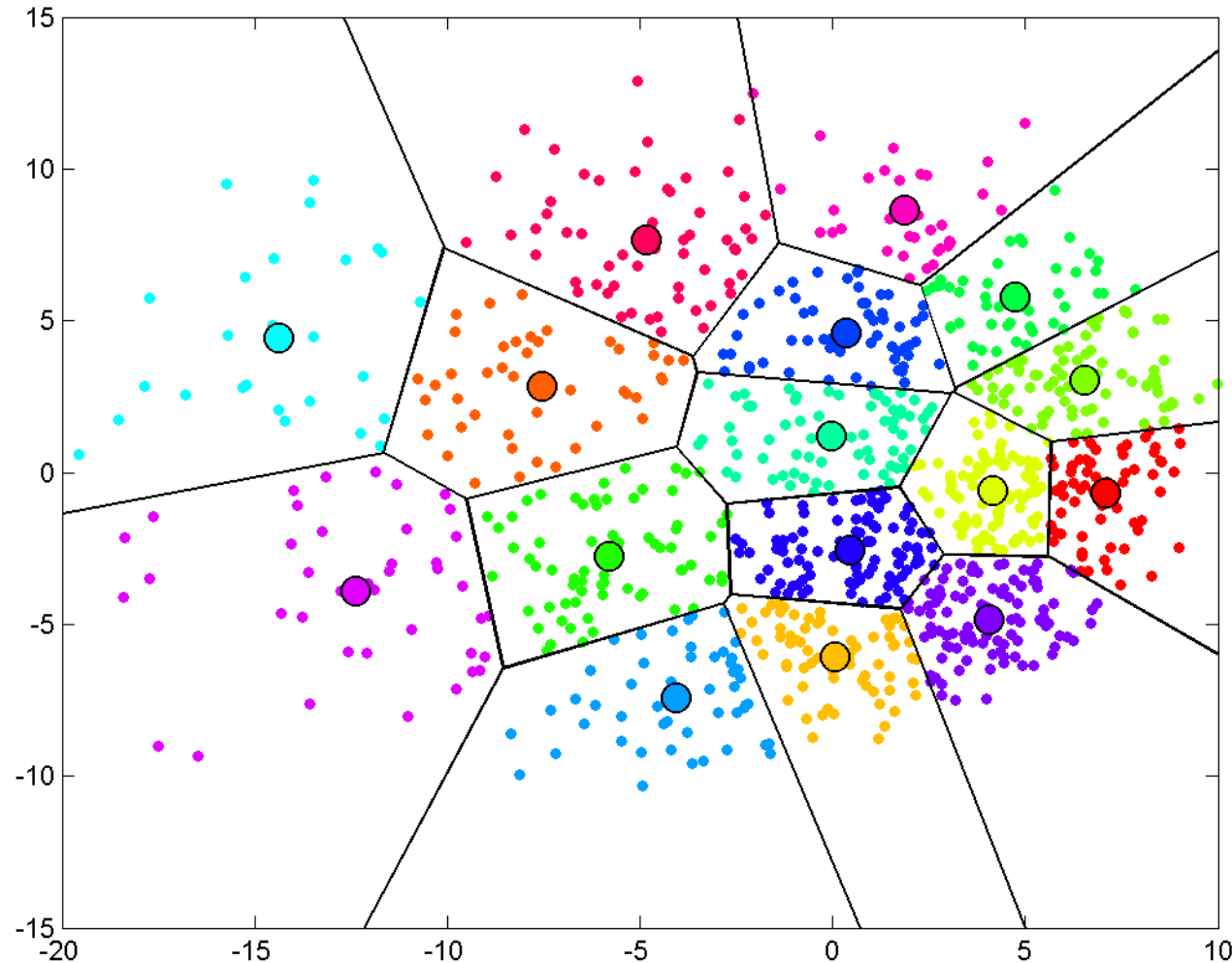
K-MEANS

For the iteration 1

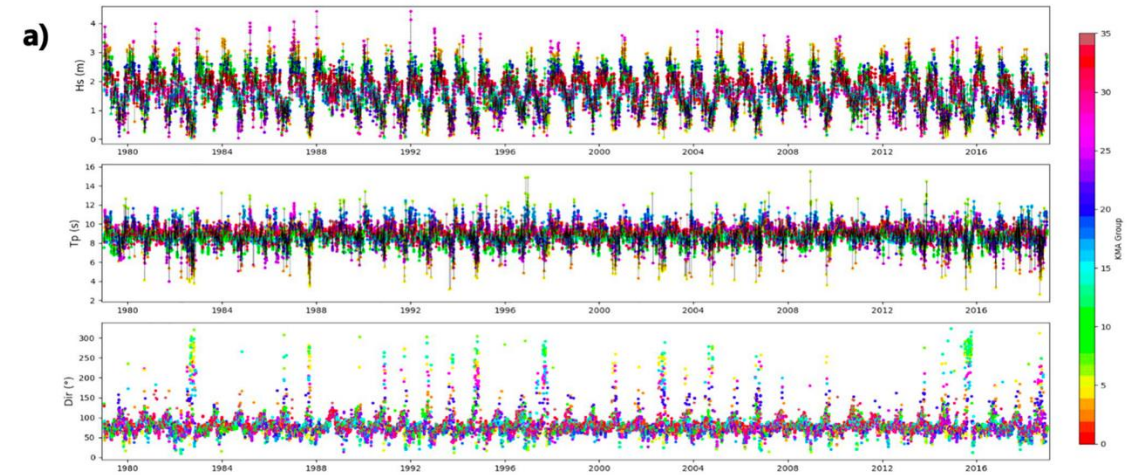
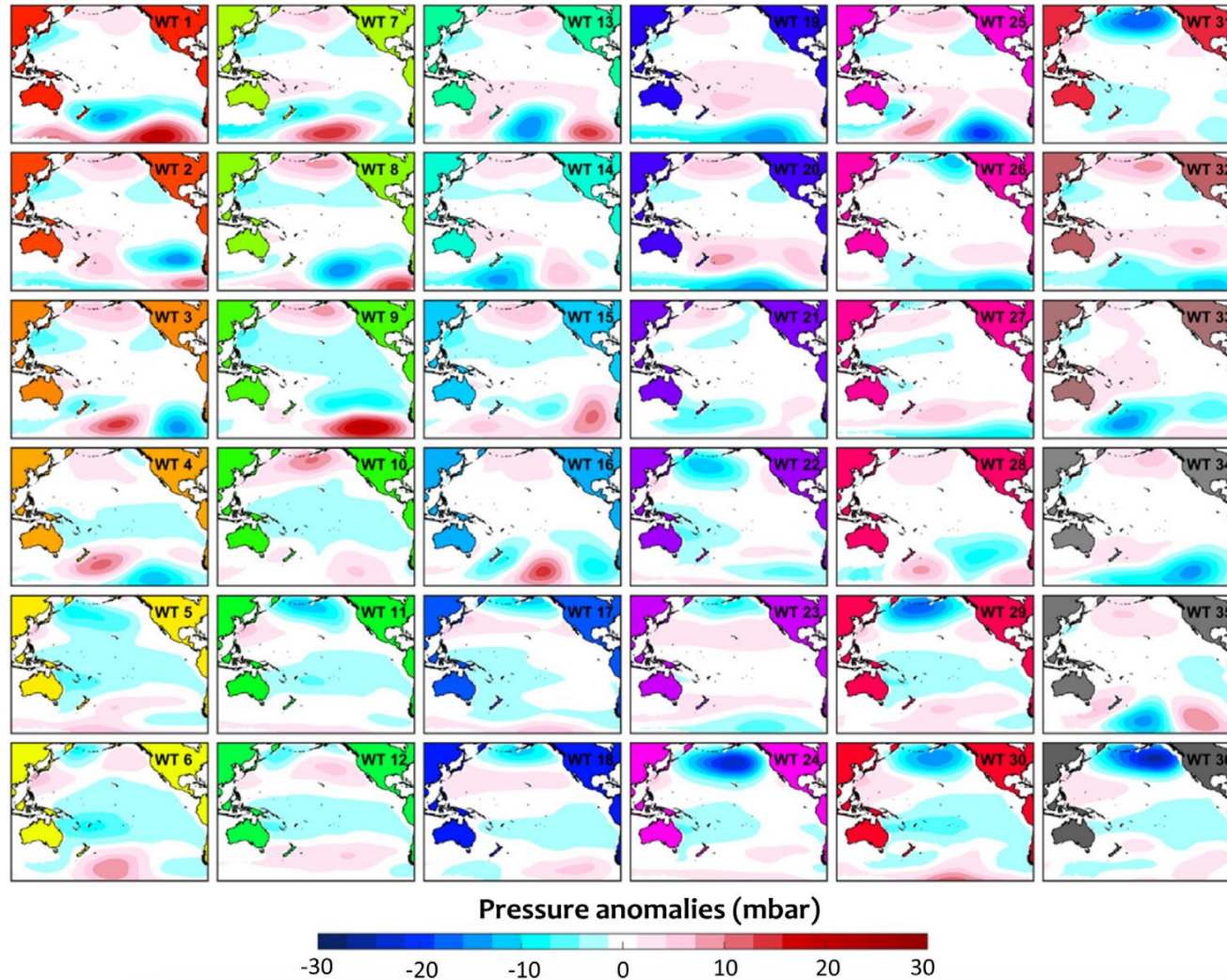
x_i → Assigned to group j
 $j = \min \left\{ \|x_i - v_j\|, j = 1, \dots, M \right\}$

→ Group j is updated from the average
of all the data corresponding to that
group

$$v_j^{r+1} = \sum_{x_i \in C_j} x_i / n_j$$

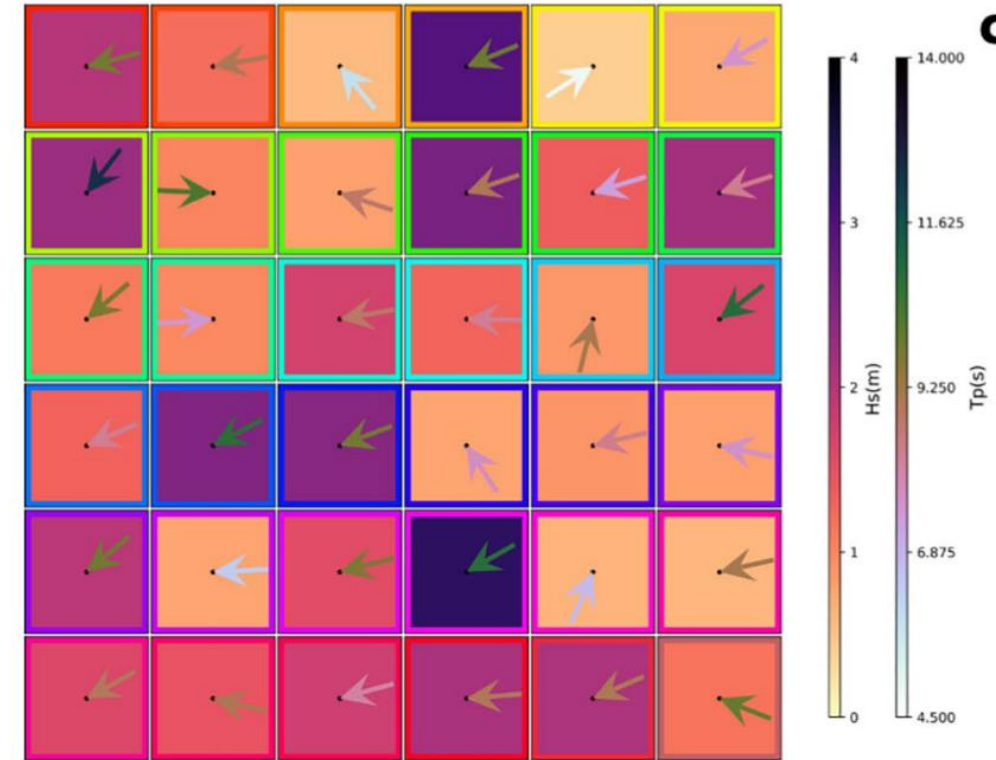
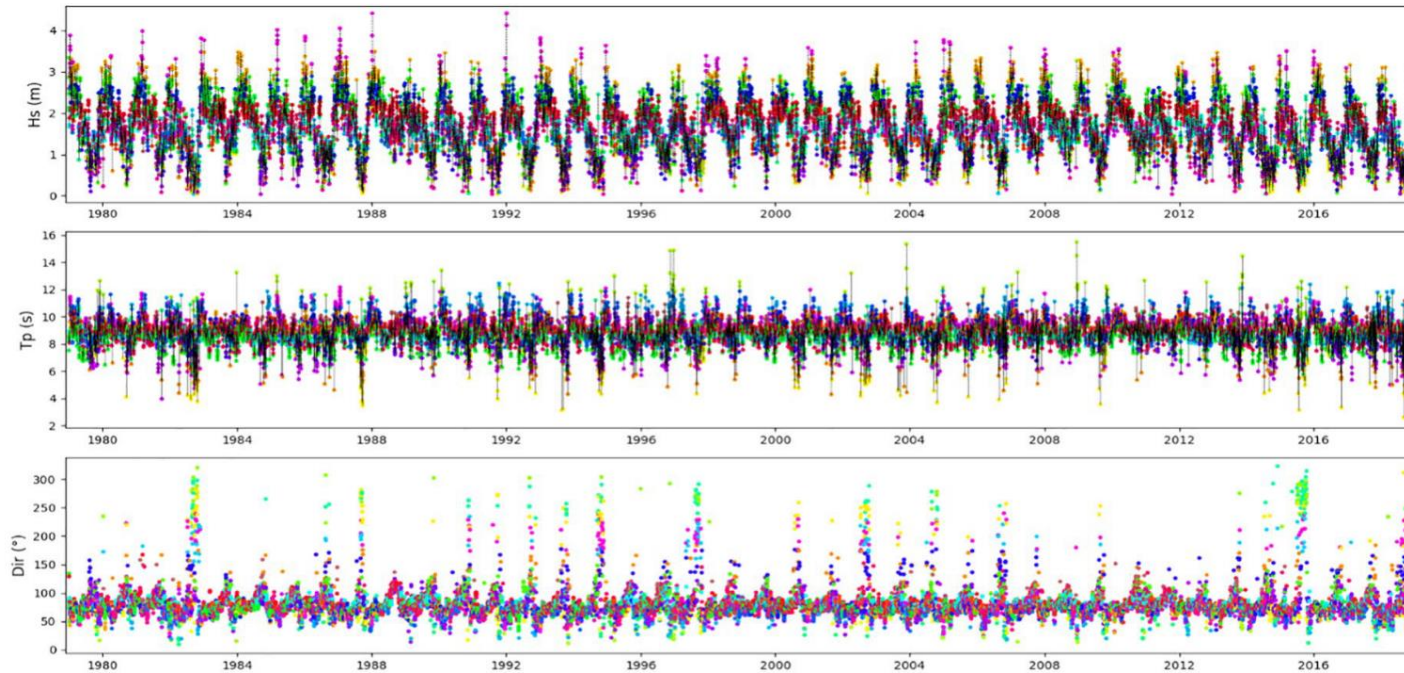


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a)



Thank you

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